Experiment 20: Planet Database Using Prolog

Aim: To write and execute a Prolog program to create a database of planets and retrieve their details.

Program :

Save the file as planets.pl.

```
% Planet Database
% planet(Name, Order_from_Sun, Type).

planet(mercury, 1, terrestrial).
planet(venus, 2, terrestrial).
planet(earth, 3, terrestrial).
planet(mars, 4, terrestrial).
planet(jupiter, 5, gas_giant).
planet(saturn, 6, gas_giant).
planet(uranus, 7, ice_giant).
planet(neptune, 8, ice_giant).

% Rules
find_planet(Name, Order, Type) :-
    planet(Name, Order, Type).

terrestrial_planet(Name) :-
    planet(Name, _, terrestrial).

gas_planet(Name) :-
    planet(Name, _, gas_giant).
```

Execution and Output

Step 1: Load the program

?- consult('C:/Users/yadal/OneDrive/Documents/ai/planets.pl').

Output:

true.

Step 2: Display all planet details

?- planet(Name, Order, Type).

Output:

Name = mercury,

Order = 1,

Type = terrestrial ;

Name = venus,

Order = 2,

Type = terrestrial ;

...

Press ; to display the next result.

Step 3: Find the position of Earth

?- find_planet(earth, Order, Type).

Output:

Order = 3,

Type = terrestrial.

Step 4: Find terrestrial planets

?- terrestrial_planet(Name).

Output:

Name = mercury ;

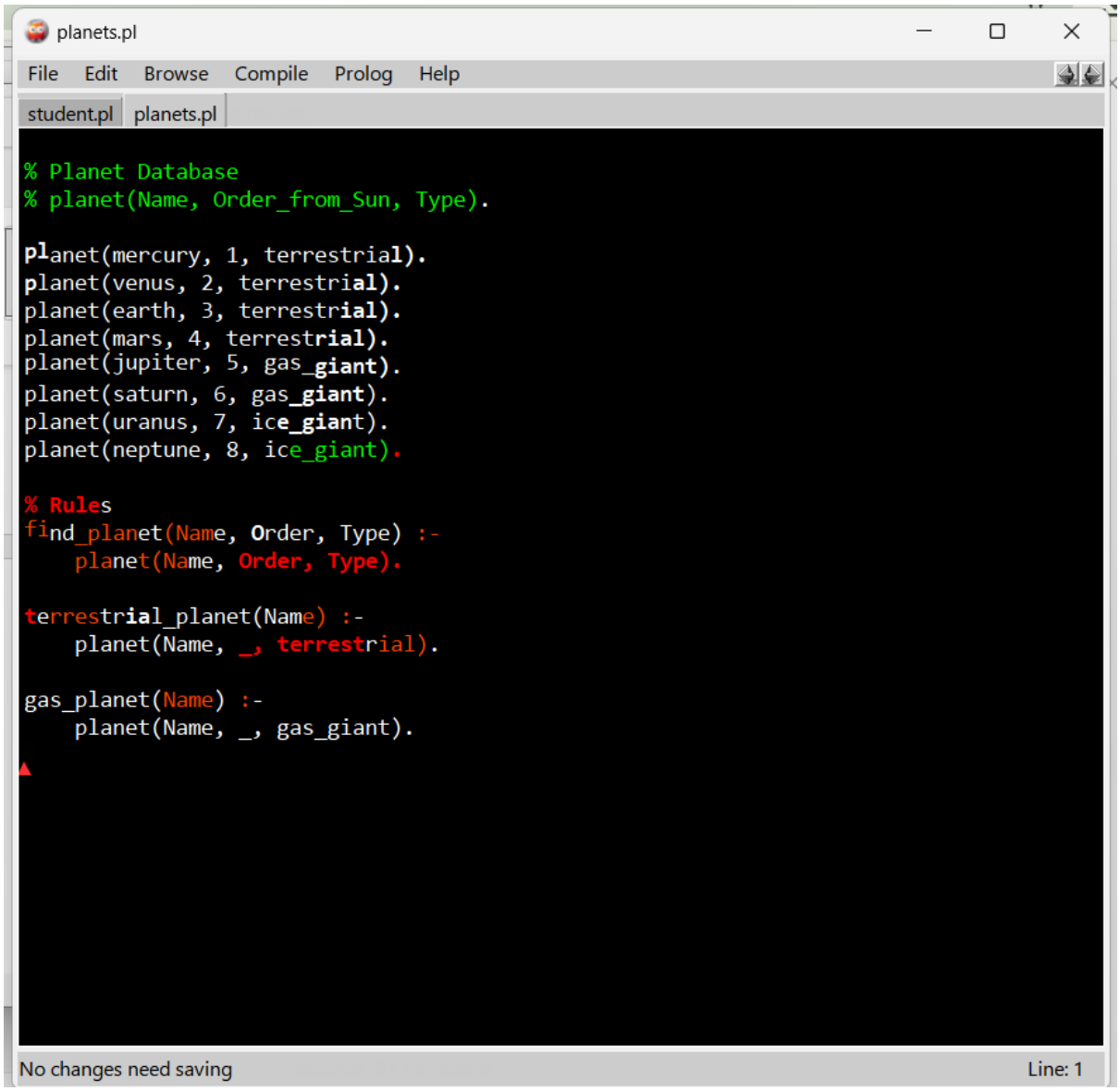
Name = venus ;

Name = earth ;

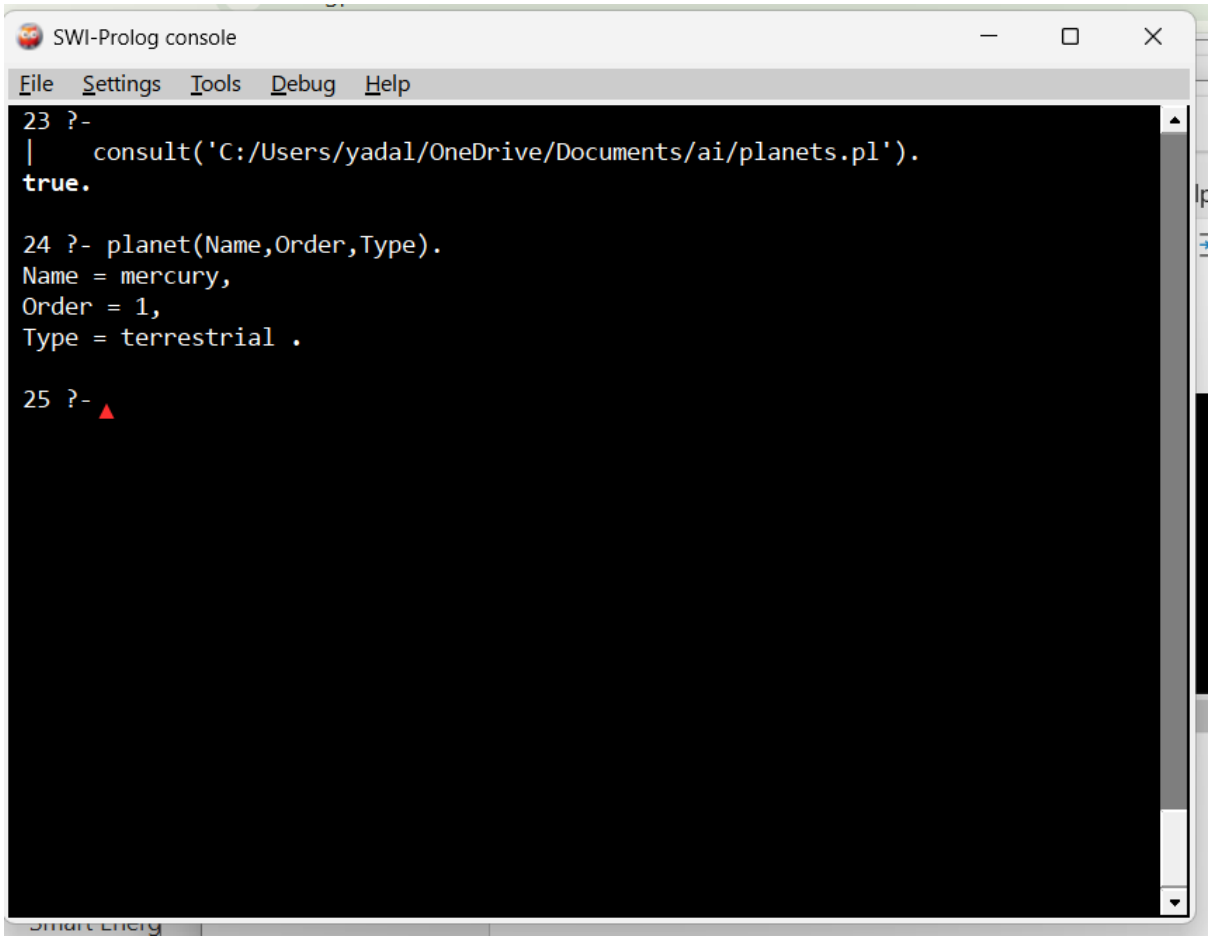
Name = mars.

Step 5: Find gas giant planets

?- gas_planet(Name).



Output:



Result

Thus, the Prolog program was successfully written and executed to create a planet database and retrieve planet names, their order from the Sun, and their types using facts and rules.